

multiplicityInClass

September 25, 2026

```
[1]: %matplotlib widget
```

```
[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.special as sp
```

```
[3]: pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', None)
```

```
[4]: def factorial(x):
    return sp.factorial(x, exact=True)

def multiplicity(N, q):
    return factorial(q+N-1)//factorial(q)//factorial(N-1)

def logArray(array):
    import math
    return [math.log(x) for x in array]
```

0.1 two einstein solids

0.1.1 3 particles each and 6 energy units

```
[5]: table0 = pd.DataFrame()
table0['q_A'] = range(0,6+1, 1)
table0['q_B'] = range(6, 0-1, -1)
table0['mult_A'] = [multiplicity(3, i) for i in table0['q_A']]
table0['mult_B'] = [multiplicity(3, i) for i in table0['q_B']]
table0['mult_total'] = table0['mult_A']*table0['mult_B']
```

```
[6]: table0
```

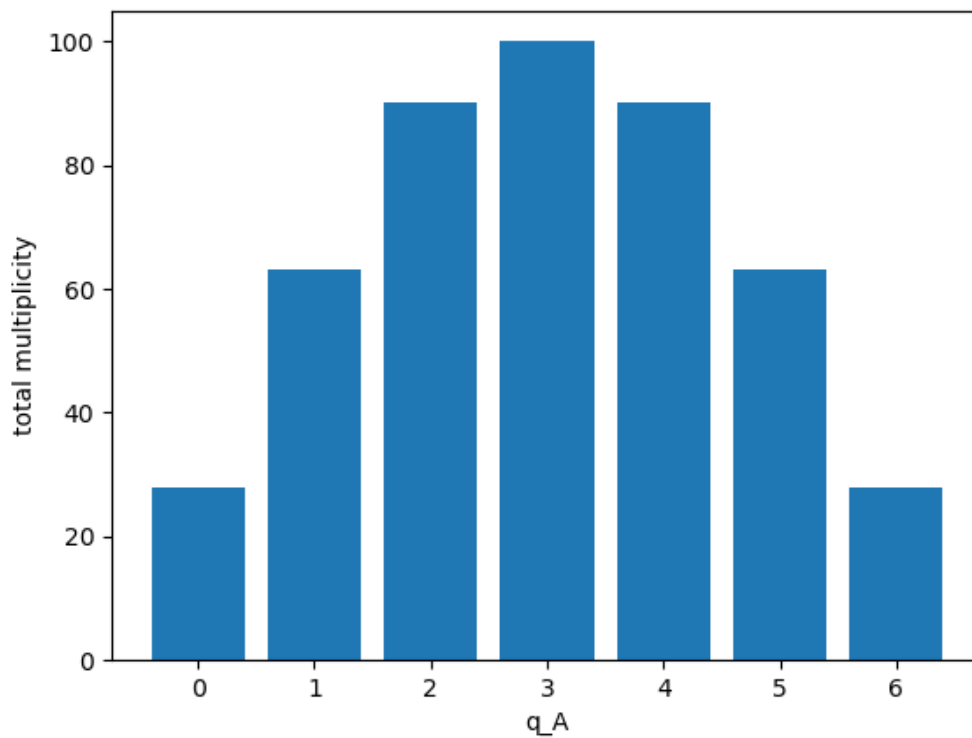
```
[6]:   q_A  q_B  mult_A  mult_B  mult_total
0    0    6        1       28         28
1    1    5        3       21         63
2    2    4        6       15         90
```

3	3	3	10	10	100
4	4	2	15	6	90
5	5	1	21	3	63
6	6	0	28	1	28

```
[6]: fig0, ax0 = plt.subplots()

ax0.bar(table0['q_A'], table0['mult_total'])
ax0.set_xlabel('q_A')
ax0.set_ylabel('total multiplicity')
```

```
[6]: Text(0, 0.5, 'total multiplicity')
```



```
[7]: table0['mult_total'].sum()
```

```
[7]: 462
```

```
[8]: multiplicity(6,6)
```

```
[8]: 462
```

```
[9]: table0.sort_values('mult_total', ascending=False).iloc[0]
```

```
[9]: q_A      3
     q_B      3
     mult_A   10
     mult_B   10
     mult_total 100
     Name: 3, dtype: int64
```

```
[10]: table0['probability'] = table0['mult_total']/table0['mult_total'].sum()
```

```
[11]: table0
```

```
[11]:
```

	q_A	q_B	mult_A	mult_B	mult_total	probability
0	0	6	1	28	28	0.060606
1	1	5	3	21	63	0.136364
2	2	4	6	15	90	0.194805
3	3	3	10	10	100	0.216450
4	4	2	15	6	90	0.194805
5	5	1	21	3	63	0.136364
6	6	0	28	1	28	0.060606

```
[12]: def multiTable(N_a, N_b, q):
      table0 = pd.DataFrame()
      table0['q_A'] = range(0,q+1, 1)
      table0['q_B'] = range(q, 0-1, -1)
      table0['mult_A'] = [multiplicity(N_a, i) for i in table0['q_A']]
      table0['mult_B'] = [multiplicity(N_b, i) for i in table0['q_B']]
      table0['mult_total'] = table0['mult_A']*table0['mult_B']
      table0['probability'] = table0['mult_total']/table0['mult_total'].sum()
      table0['S_A'] = logArray(table0['mult_A'])
      table0['S_B'] = logArray(table0['mult_B'])
      table0['S_total'] = logArray(table0['mult_total'])
      return table0
```

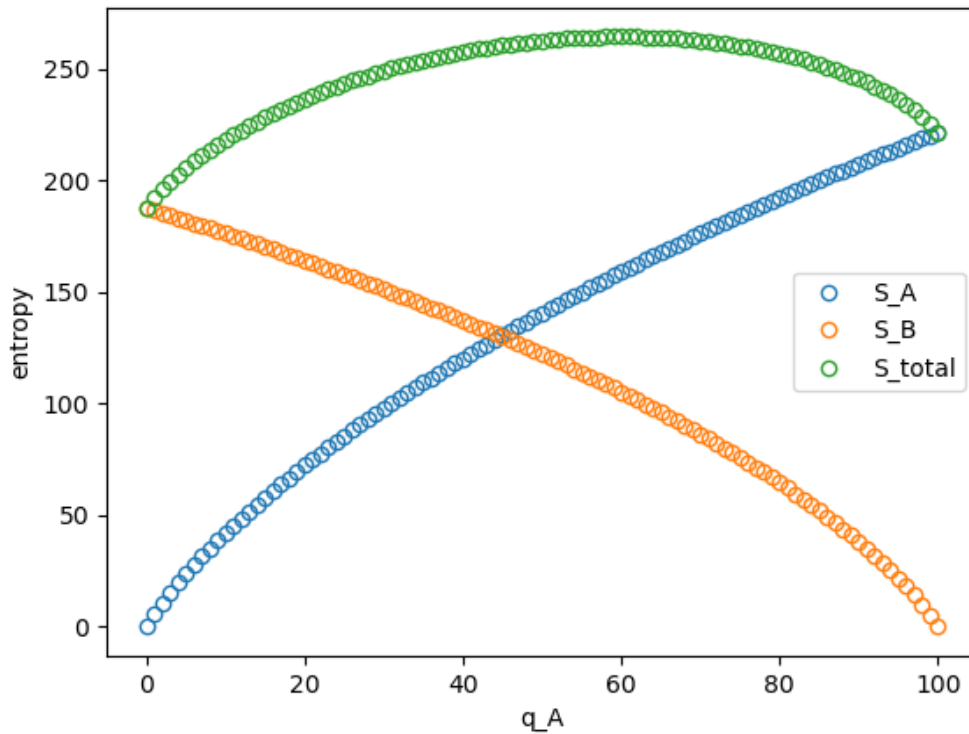
```
[13]: table1 = multiTable(300, 200, 100)
```

```
[14]: fig1, ax1 = plt.subplots()

      ax1.plot(table1['q_A'], table1['S_A'], 'o', mfc='None')
      ax1.plot(table1['q_A'], table1['S_B'], 'o', mfc='None')
      ax1.plot(table1['q_A'], table1['S_total'], 'o', mfc='None')

      ax1.set_xlabel('q_A')
      ax1.set_ylabel('entropy')
      ax1.legend(['S_A', 'S_B', 'S_total'])
```

```
[14]: <matplotlib.legend.Legend at 0x7bbeed5166c0>
```



```
[17]: table1['mult_total'].sum()
```

```
[17]: 92617601588848787700794369127348784356121334262853897042393434278491595498450019
668972021860619724231355309579374480
```

```
[18]: multiplicity(500,100)
```

```
[18]: 92617601588848787700794369127348784356121334262853897042393434278491595498450019
668972021860619724231355309579374480
```

```
[20]: import math
```

```
[21]: math.log(multiplicity(500,100))
```

```
[21]: 267.0231798068631
```

```
[16]: fig2, ax2 = plt.subplots()
```

